





Design, and Implementation of Autonomous System For Path Identification, and Robotic Execution, with 3D Vision, and ROS2

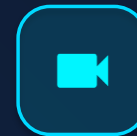
6-Axis Robotic Arm with Closed-Loop Control , and Vision System



MCU



Control System



Machine Vision

Team Members: [Ahmed Al-Duais]

PROJECT OVERVIEW

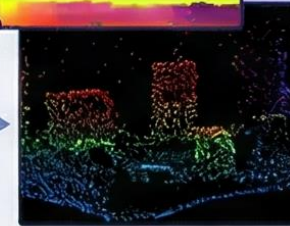
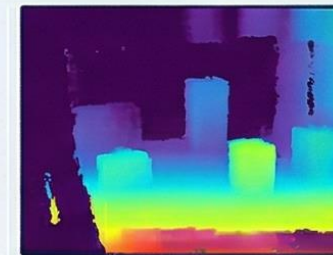
Advanced 6-axis robotic system with embedded intelligence



3D vision, and Depth sensing

Determining The Aspects of Path in 3D Dimension

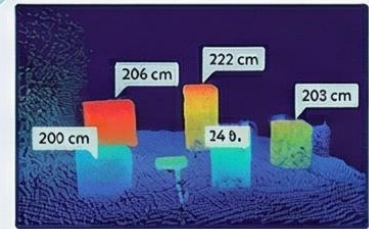
3D VISION & DEPTH SENSING



Point
Cloud



Depth Sensor



- 3D Vision & Depth Sensing

PROJECT OVERVIEW

Advanced 6-axis robotic system with embedded intelligence



3D vision, and Depth sensing

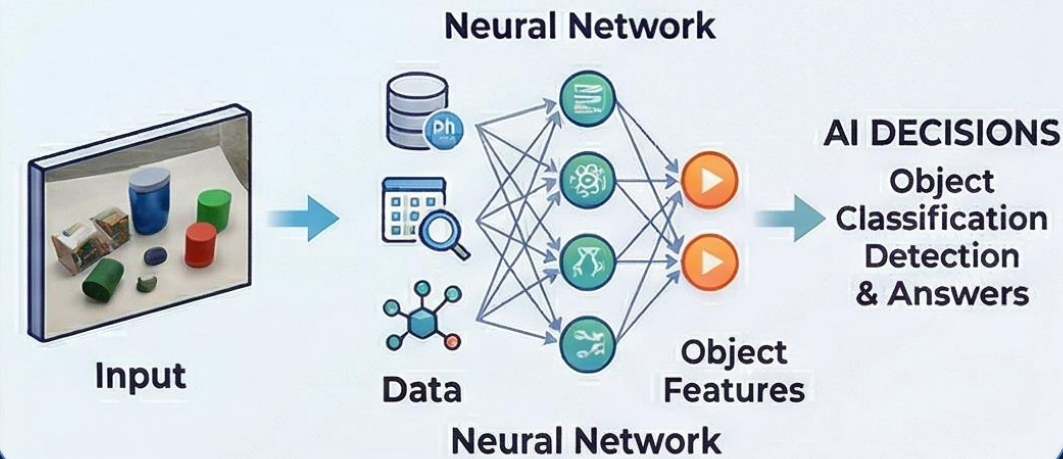
Determining The Aspects of Path in 3D Dimension



Image Processing, AI

Processing The Taken Image, to Identify The Path of the Operation

IMAGE PROCESSING & AI



- Image Processing

PROJECT OVERVIEW

Advanced 6-axis robotic system with embedded intelligence



3D vision, and Depth sensing

Determining The Aspects of Path in 3D Dimension



Image Processing, AI

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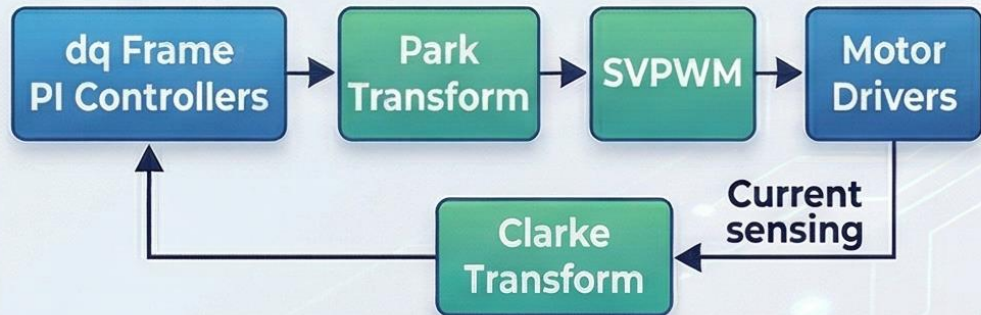


Control System

Specify the Parameters Needed to Execute The Identified Path

FIELD CONTROL SYSTEM

COMMAND GENERATION



CLOSED-LOOP FEEDBACK

- Control System

PROJECT OVERVIEW

Advanced 6-axis robotic system with embedded intelligence



3D vision, and Depth sensing

Determining The Aspects of Path in 3D Dimension



Image Processing, AI

Processing The Taken Image, to Identify The Path of the Operation



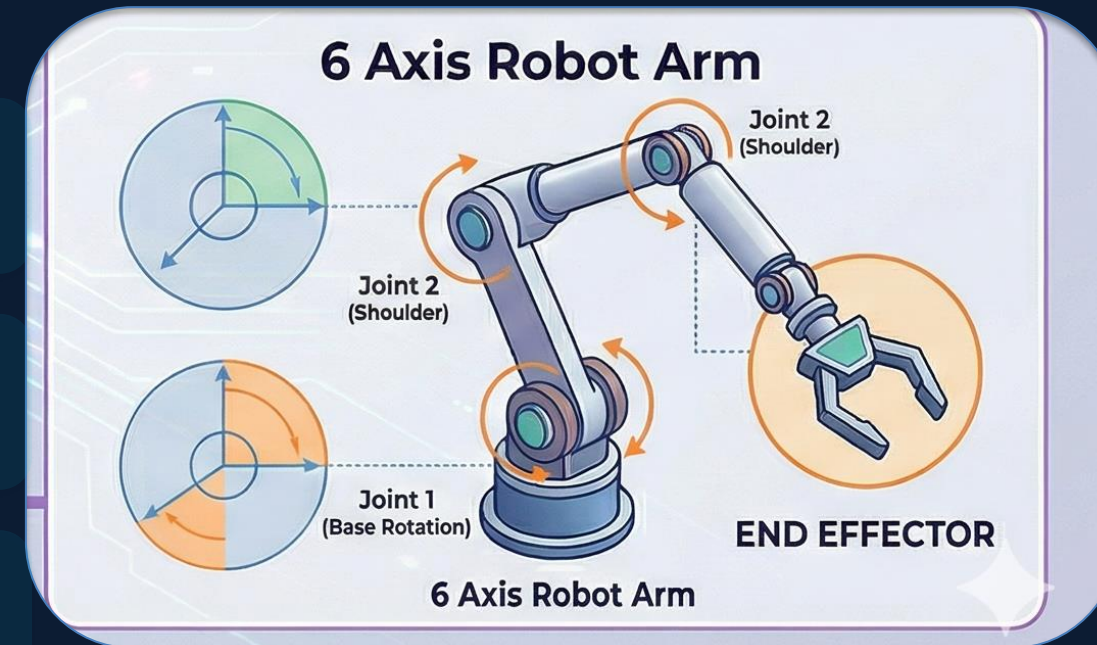
Control System

Specify the Parameters Needed to Execute The Identified Path



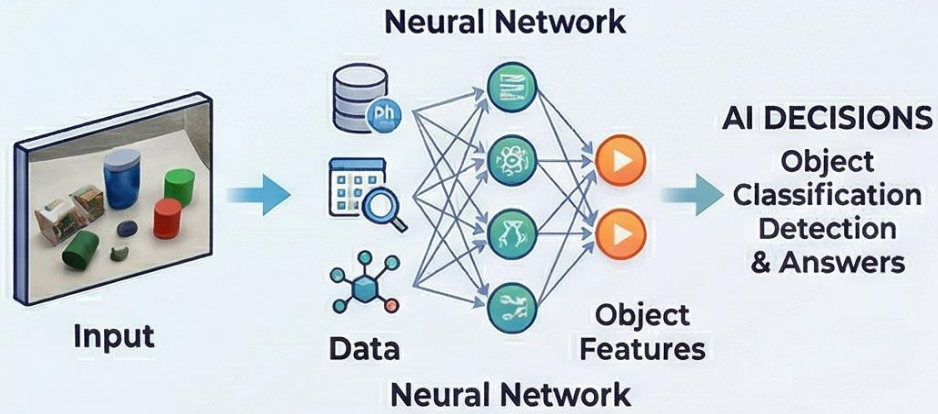
6 Axis Robotic Arm

Execute The Identified Path

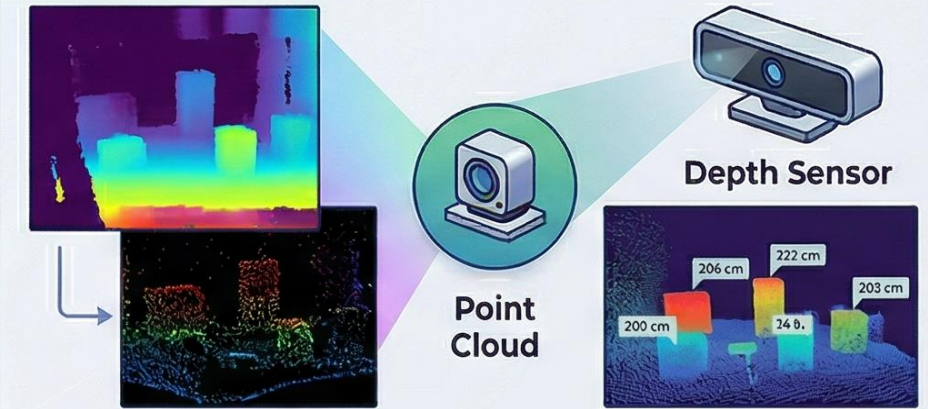


- 6Axis Robot

IMAGE PROCESSING & AI



3D VISION & DEPTH SENSING

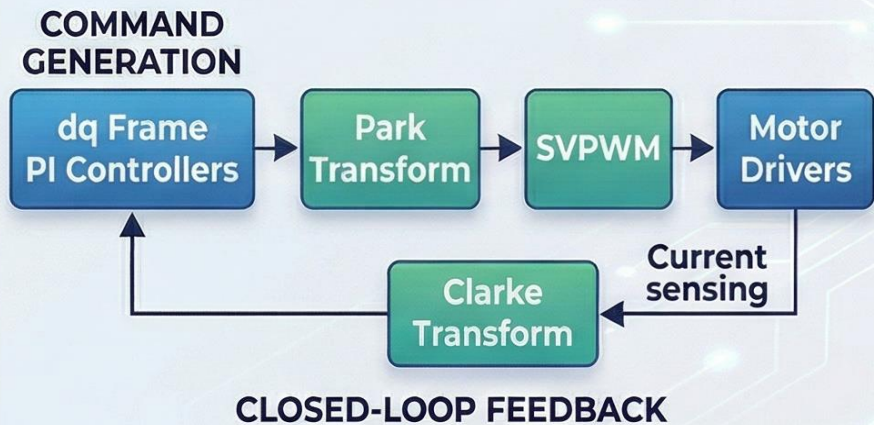


DATA
ACQUISITION

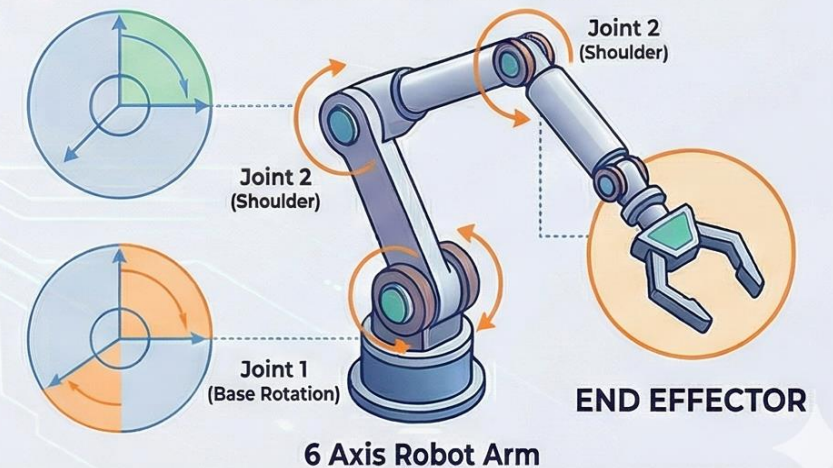


6 AXIS ROBOT ARM

FIELD CONTROL SYSTEM



6 Axis Robot Arm



MOTION FEEDBACK

